Interacting-Enhancing Feature Transformer for Cross-Modal Remote-Sensing Image and Text Retrieval

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***Abstract*— Cross-modal remote-sensing image–text retrieval (CMRSITR) is a challenging topic in the remote-sensing (RS) community. It has gained growing attention because it can be flexibly used in many practical applications. In the current deep era, with the help of deep convolutional neural networks (DCNNs), many successful CMRSITR methods have been proposed. Most of them first learn valuable features from RS images and texts, respectively. Then, the obtained visual and textual features are mapped into a common space for the final retrieval. The above operations are feasible; however, two difficulties are still to be solved. One is that the semantics within the visual and textual features are misaligned due to the independent learning manner. The other one is that the deep links between RS images and texts cannot be fully explored by simple common space mapping. To overcome the above challenges, we propose a new model named interacting-enhancing feature transformer (IEFT) for CMRSITR, which regards the RS images and texts as a whole. First, a simple feature embedding module (FEM) is developed to map images and texts into the visual and textual feature spaces. Second, an information interacting-enhancing module (IIEM) is designed to simultaneously model the inner relationships between RS images and texts and enhance the visual features. IIEM consists of three feature interacting-enhancing (FIE) blocks, each of which contains an intermodality relationship interacting (IMRI) subblock and a visual feature enhancing (VFE) subblock. The duty of IMRI is to exploit the hidden relations between cross-modal data, while the responsibility of VFE is to improve the visual features. By combining them, semantic bias can be mitigated, and the complex contents of RS images can be studied. Finally, the retrieval module (RM) is constructed to generate the matching scores for deciding the search results. Extensive experiments are conducted on four public RS datasets. The positive results demonstrate that our IEFT can achieve superior retrieval performance compared with many existing methods. Our source codes are available at https://github.com/TangXu-Group/Cross-modal-remote-sensing-image-and-text-retrieval-models/tree/main/IEFT.**

***Index Terms*— Cross-modal remote-sensing image–text retrieval (CMRSITR), transformer.**

I. Introduction

With the development of Earth observation technologies, collecting a large number of remote-sensing (RS) images is not a difficult task anymore [1], [2], [3]. They provide new opportunities for scholars to explore our planet. Before using the proper interpretation methods to mine the information from them, managing these large RS images according to specific demands is an indispensable step [4]. To this end, information retrieval entries the RS community. There are two common retrieval techniques for RS images, that is, content-based RS image retrieval (CBRSIR) [5], [6], [7] and cross-modal RS image–text retrieval (CMRSITR) [8], [9], [10], [11]. CBRSIR aims to search the target RS images with similar contents to the query RS image, highlighting the image contents' understanding. CMRSITR refers to using one modality of data to find the most semantic consistency data in another modality, which emphasizes the exploration of the latent correspondence between images and texts. Each of them has its advantages. However, compared with CBRSIR, CMRSITR (the topic of this article) attracts more attention recently since it establishes a bridge between the RS image understanding and language comprehension, which is beneficial to many practical applications [12], [13], [14], [15].

The study of CMRSITR is still in its infancy. With the help of deep-learning techniques, several CMRSITR methods have been proposed [8], [9], [10], [16], [17]. Most of them are dual-branch networks. One branch aims at learning the visual features from RS images, and the other focuses on extracting the textual representations from texts. When the different features are obtained, some specific layers embedded at the top of the model are developed to map them into the joint space so that the similarities between RS images and texts can be measured directly. Here, the common tools for visual feature learning are deep convolutional neural networks (DCNNs) [18], the usual techniques for textual representation learning are recurrent neural networks (RNNs) [19], and the popular loss functions are the triplet loss and its variants [20]. Although these dual-branch methods are feasible, their

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performance cannot still reach what we expected. The reasons behind this are twofold. First, since those models learn the features of RS images and texts independently, in general, the semantic gaps between different feature representations are distinct [21]. In other words, the semantics within visual features and textual representations are misaligned [22], which would reduce the reliability of the resemblance measurement. Second, the above CMRSITR methods mainly emphasize feature learning. However, modeling relations between RS images and texts properly is often ignored. This is adverse to mining the semantic correspondence under this cross-modal scenario [23].

The transformer [24], [25], [26] has recently drawn scholars' attention because of its inherent advantages in modeling relationships hidden in the input data. The transformer-based models become prevalent in diverse computer vision applications [27], [28], [29], as well as the cross-modal image–text retrieval [22], [23], [30]. On the one hand, owing to the peculiar structure, the images and texts are fed into a unified network, and the corresponding visual features and textual representations can be learned simultaneously. This guarantees semantic consistency between different features. On the other hand, benefiting from the self-attention mechanism (SAM), the relations between images and texts can also be modeled deeply in the transformer-based models. This is conducive to establishing the semantic correspondence between two modalities effectively. One question is raised, can we transfer these methods to RS tasks directly?

The answer is probably negative. Although the above-mentioned methods have shown their superiorities in natural image processing, there are still some challenges to be addressed when applying them to the RS images. First, the transformer-based methods focus on building cross-modal relationships. However, they do not attach importance to image feature learning. This limitation is inconspicuous to natural images but is catastrophic to RS images since they contain a large number of land covers with diverse types and multiple scales. Second, the images' spatialwise information is fully considered when they apply SAM to model the cross-modal relationships. Nevertheless, the channelwise knowledge hidden in the corresponding feature maps does not get the attention it deserves, limiting their ability to understand RS images.

To address the above problems, inspired by vision-and-language transformer (ViLT) [23], we propose a new CMRSITR method named interacting-enhancing feature transformer (IEFT). The IEFT contains a feature embedding module (FEM), an information interacting-enhancing module (IIEM), and a retrieval module (RM). FEM introduces a pretrained Bert tokenizer and a simple linear projector to transform texts and RS images into feature embeddings. Then, these embeddings are fed into IIEM simultaneously to keep feature consistency. IIEM is developed based on the transformer. On the one hand, it leverages SAM to model the relationships between texts and RS images at the spatial level. On the other hand, considering the complex contents hidden in RS images, IIEM also improves the visual feature embeddings by investigating their channelwise and localwise information. RM comprises three transformer encoders and an image–text matching (ITM) head. The transformer encoders aim to discover inner relations between texts and RS images. The ITM head focuses on generating the matching scores, which can be used directly to get the retrieval results.

1) We propose an end-to-end CMRSITR model (i.e., IEFT). Instead of exploring valuable information from RS images and texts individually, IEFT simultaneously processes RS images and texts, so the visual and textual knowledge can be learned with small distribution misaligned.
2) IEFT can inherently explore the relationships between RS images and texts by taking advantage of SAM within the transformer. In this way, the final cross-modal matching decisions tend to be creditable.
3) To further mine the useful information, we develop two simple, yet useful units to capture the complex contents within RS images from the channel and local aspects. By embedding them into our IEFT, the obtained visual features for RS images are more discriminative.
4) Extensive experiments are conducted on four benchmark datasets, and the encouraging results demonstrate that the proposed method is effective.

The rest of this article is organized as follows. Previous works related to CMRSITR are reviewed in Section II. Then, the proposed method is introduced in Section III. The experimental results and discussion are given in Section IV. Finally, the conclusions are summarized in Section V.

II. Related Work

A. Cross-Modal Image–Text Retrieval

Cross-modal image–text retrieval refers to finding semantically similar data between images and texts. According to the type of images, we review some published works from the following two aspects, including cross-modal natural scene image-text retrieval (CMNSITR) and CMRSITR.

1) Cross-Modal Natural Scene Image–Text Retrieval: CMNSITR is always a hot research topic in image processing, which aims to find semantically similar data across images and texts [31], [32]. Especially in the current deep era, the rapid development of CMNSITR is obvious to all [33]. As early as 2018, You et al. [34] developed an end-to-end semantic embedding model for CMNSITR. This method leverages DCNNs to learn hierarchical features from both images and texts first. Then, the local information (within the middle-level features) and the semantic messages (within the high-level features) are combined by a specific learning objective function to complete the cross-modal retrieval. The positive experimental results confirm its helpfulness. This kind of solution can also be found in [35], [36], and [37]. Although the above methods' performance is good, they still face the problem of heterogeneity gaps. To mitigate this issue, Gu et al. [32] developed a generative model to map images and texts into a common semantic space. By introducing the generative adversarial scheme, the global abstract and local grounded features can be explored from dual-modal data. Ji et al. [38] developed two specific networks based on DCNN and the

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long short-term memory (LSTM) model. After fully mining the valuable information from images and texts using two networks, the different information is mapped into a common space for completing the cross-modal similarity matching. Considering the diverse contents within images, some scholars add object detection tools into their cross-modal method, such as [39], [40], [41], [42], and [43]. This way, the images can be interpreted comprehensively so that the cross-modal retrieval performance can be enhanced.

Recently, when the transformer appears, it attracts researchers' attention instantly due to its strong capacity for relationship learning. In the CMNSITR field, many interesting works have been proposed and achieved impressive results. For instance, vision-and-language BERT (ViLBERT) [44] combines three basic transformers to deal with the cross-modal retrieval tasks. The first two transformers are used to learn the features from images and texts separately. Here, rather than a whole image, the various objects within the image are detected for visual feature learning. The last one is used to model the inner relationships for measuring the similarities. Meanwhile, universal image-text representation (UNITER) [30] and object-semantics aligned pre-training (OSCAR) [22] directly introduce a single transformer to both process images (at the object level) and texts and study the cross-modal relationships. To further ensure the stability and discrimination of visual features, ViLT [23] adds a lightweight learnable linear projector to map whole images into feature embeddings. Then, a transformer is adopted to analyze the cross-modal relations. Unlike the methods mentioned above, align the image and text representations before fusing (ALBEF) [45] uses web data and a momentum distillation scheme to enhance the retrieval performance. This approach improves the model's ability to understand and reason about cross-modal relationships.

2) Cross-Modal Remote-Sensing Image–Text Retrieval: In the CMRSITR community, some successful methods have been proposed. Yuan et al. [16] developed an asymmetric multimodal feature matching network (AMFMN) for CMRSITR tasks, in which the multiscale characteristic of RS images are fully considered for deeply interpreting their diverse land covers. AMFMN first introduces the multiscale visual self-attention scheme to find the salient areas within RS images and guide the text representation learning at the same time. Then, a specific triplet loss is developed to bridge the relations between data from different modalities. The positive results counted on their constructed dataset imply the effectiveness of AMFMN. To enhance the feasibility of the retrieval approach, a lightweight retrieval model was presented in [8], in which the multiscale feature learning scheme, abundant feature filter mechanism, lightweight text representation extraction strategy, and knowledge distillation policy are combined for CMRSITR. Meanwhile, another model was published by the same group [10]. In this model, the authors emphasize the importance of visual feature learning in CMRSITR. Both the global and local information are considered when transforming RS images into discriminative visual features. Such that the complex contents hidden in RS images can be understood for cross-modal retrieval. The above methods are supervised, whose behavior depends on many labeled data. To relieve this burden, Mikriukov et al. [46] introduced an unsupervised method for CMRSITR, where contrastive learning, hashing, and multiobjective optimization techniques are integrated. The encouraging results confirm its helpfulness. Despite the advancements in implementing these methods, a significant opportunity for enhancement persists. Because the current approach primarily utilizes triplet loss to regulate the semantic associations between different modalities, ignoring the challenging situation that the feature spaces of each modality are distinct and independent.

B. Transformer-Based Methods for RS Data

Transformer [24] was first proposed for natural language processing [25], [47], [48], [49]. Inspired by its powerful ability in long-range dependence learning, the transformer has become prevalent in computer vision [50], as well as the RS community. Bazi et al. [51] first applied a transformer to RS image processing. They leveraged vision transformer (ViT) [26] to handle the RS scene classification tasks. Owing to the multihead attention (MHA) mechanisms, the global contextual information can be fully explored so that the classification results are competitive. Another transformer-based RS scene classification method was proposed in [52]. Based on ViT, the authors developed specific schemes to capture the spatialwise and channelwise information from RS scenes. In this way, the detailed geometric clues and the contributions of multiple feature maps can be utilized to distinguish the various scenes. Ma et al. [53] proposed a homo-heterogeneous transformer learning (HHTL) framework for the RS scene classification. This method fully considers the homogenous and heterogeneous knowledge within RS scenes. Along with the transformer structure, the learned features are discriminative for classification. To reduce the time costs caused by SAM, an efficient multiscale transformer was proposed in [54]. By changing the MHA mechanism into a 2-D version and introducing the dilated convolution, the rich contextual information within RS scenes can be explored at a small time cost.

The above transformer-based methods were scene-oriented. Many models have been developed based on the transformer for objectwise and pixelwise applications. For instance, Wang et al. [55] combined DCNN and transformer to adaptively capture local and global information from RS images so that the objects can be detected from the complex background. In this method, an adjacent feature-aligned module is developed to align adjacent features, and an interactive guidance loss is formulated to achieve joint region and boundary detection. Therefore, diverse objects can be effectively discovered. A transformer-based method named SpectralFormer was introduced in [56] for hyperspectral image classification tasks. SpectralFormer uses groupwise spectral embedding to learn local information and adopts the cross-layer adaptive fusion scheme to combine the valuable information from multiple layers. The promising results counted on several hyperspectral images confirm its effectiveness. Li et al. [57] proposed an end-to-end hybrid transformer model (TransUNetCD) for RS image change detection under the Unet framework. TransUNetCD first uses Siamese DCNN to fully learn the shallow

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and high deep features from bitemporal RS images. Then, a transformer module is developed to mine the global context information. The encouraging results reported in the literature illustrate that the transformer is useful in RS change detection. A wide-context network was introduced in [58] for RS image semantic segmentation. The detailed local information and the global contextual relations can be captured by combining the normal convolution and transformer. Thus, satisfactory semantic segmentation results can be obtained based on this luxuriant knowledge.

III. Proposed Method

A. Overall Framework

The architecture of IEFT is shown in Fig. 1, consisting of an FEM, an IIEM, and an RM. Suppose we have a set of RS image and text pairs (V, T), where $V = \{v_1, \ldots, v_N\}$ and $T = \{t_1, \ldots, t_N\}$ indicate RS images and texts, and N denotes the volume of the set. Their relationship labels are $\mathbf{m} = \{m_1, \ldots, m_N\}$, where $m_i = 1$ means that v_i and t_i contain similar concepts, while $m_i = 0$ implies that they have different concepts. FEM aims to map RS image and text pairs (V, T) into d-dimensional feature embeddings $\mathcal{F} = \{\mathbf{F}_1, \ldots, \mathbf{F}_N\}$ and $\mathcal{G} = \{\mathbf{G}_1, \ldots, \mathbf{G}_N\}$. The goals of IIEM are twofold. One is to model the intermodality relationships by mutually investigating the features corresponding to two modalities. The other is to enhance the RS image features by exploring channelwise information and salient local areas. Finally, RM continues to further fuse two modalities features for narrowing the possible semantic gaps and leverages the ITM head to output the matching scores. In the training phase, we adopt the cross-entropy loss function to optimize our model by mining the differences between the output scores and the relationship labels. In the inference phase, the matching scores are utilized to produce the retrieval results. Now, we explain each module in detail.

B. Feature Embedding Module

As illustrated in Fig. 2, FEM contains a text embedding block and an image embedding block. They are developed to transform texts and the corresponding RS images into d-dimensional features.

1) Text Embedding Block: Unlike many existing works [8], [10], we employ a pretrained Bert tokenizer [25], [44] instead of common RNNs to learn the texts' features. This is because Bert is a transformer-based model, which is consistent with the rest of our IEFT. Note that the common mask-maker strategy in Bert's training is excluded to ensure the textual integrity of our task. Also, the Bert tokenizer's parameters would be updated along with the training of the proposed IEFT.

We assume that a text t_i contains l words, which can be expressed as $t_i = \{w_i^1, \ldots, w_i^l\}$. First, a predefined vocabulary is introduced to transform words within t_i into one-hot codes $\mathbf{O}_i = [\mathbf{o}_i^1, \ldots, \mathbf{o}_i^l] \in \mathbb{R}^{d_o \times l}$, where $\mathbf{o}_i^j$ indicates the jth word's one-hot representation and its dimension is d_o. Then, these codes would be converted into the textual features $\mathbf{G}_i^E \in \mathbb{R}^{d \times l}$ by a learnable mapping matrix $\mathbf{E}^{\text{text}} \in \mathbb{R}^{d \times d_o}$, which is formulated as

$$\mathbf{G}_i^E = \mathbf{E}^{\text{text}}(\mathbf{O}_i) = \left[\mathbf{a}_i^1, \ldots, \mathbf{a}_i^l\right] \tag{1}$$

where $\mathbf{a}_i^j \in \mathbb{R}^{d \times 1}$ denotes the jth word's textual feature. Like Bert, the obtained textual feature $\mathbf{G}_i^E$ would be attached with a fixed text classification token $\mathbf{a}^0 \in \mathbb{R}^{d \times 1}$ in this article. Also, to eliminate the disturbances caused by the same words, the text positional matrix $\mathbf{G}_i^{\text{pos}} \in \mathbb{R}^{d \times (1+l)}$ is added to $\mathbf{G}_i^E$ for enriching the textual features. The above operations can be defined as

$$\mathbf{G}_i' = \left[\mathbf{a}^0, \mathbf{a}_i^1, \ldots, \mathbf{a}_i^l\right] + \mathbf{G}_i^{\text{pos}}. \tag{2}$$

Finally, the textual embeddings $\mathbf{G}_i = [\mathbf{g}^0, \mathbf{g}_i^1, \ldots, \mathbf{g}_i^l] \in \mathbb{R}^{d \times (1+l)}$ can be obtained by adding $\mathbf{G}_i'$ and the text-type identification matrix $\mathbf{G}_i^{\text{type}}$, that is,

$$\mathbf{G}_i = \mathbf{G}_i^{\text{type}} + \mathbf{G}_i'. \tag{3}$$

Here, we set the token of $\mathbf{G}_i^{\text{type}}$ to be "0."

2) Image Embedding Block: Inspired by ViT [26], we develop a lightweight linear projector to generate the RS image embeddings. For an RS image v_i whose height, width, and channel are H, W, and C, respectively, we first slice and flatten it into n patches $P_i = \{p_i^1, \ldots, p_i^n\} \in \mathbb{R}^{(h \times w \times C) \times n}$, where h and w denote the heights and widths of patches, respectively, and their values are the same in general. Then, a learnable linear projection matrix $\mathbf{E}^{\text{image}} \in \mathbb{R}^{d \times (h \times w \times C)}$ is designed to map patches into d-dimensional visual features $\mathbf{F}_i^E \in \mathbb{R}^{d \times n}$. The formulation of this step is

$$\mathbf{F}_i^E = \mathbf{E}^{\text{image}}(P_i) = \left[\mathbf{b}_i^1, \ldots, \mathbf{b}_i^n\right] \tag{4}$$

where $\mathbf{b}_i^j \in \mathbb{R}^{d \times 1}$ means the jth RS patches' visual feature. Next, we concatenate $\mathbf{F}_i^E$ with an image classification token $\mathbf{b}_i^0 \in \mathbb{R}^{d \times 1}$ and add the image positional embedding vector $\mathbf{F}_i^{\text{pos}} \in \mathbb{R}^{d \times (1+n)}$ to improve the visual features with the image structure information. This operation can be defined as

$$\mathbf{F}_i' = [\mathbf{b}_i^0, \mathbf{b}_i^1, \ldots, \mathbf{b}_i^n] + \mathbf{F}_i^{\text{pos}}. \tag{5}$$

Finally, we can get the RS image embeddings $\mathbf{F}_i = [\mathbf{f}^0, \mathbf{f}_i^1, \ldots, \mathbf{f}_i^n] \in \mathbb{R}^{d \times (1+n)}$ by combining the enhanced visual features $\mathbf{F}_i'$ and the image-type identification matrix $\mathbf{F}_i^{\text{type}}$, that is,

$$\mathbf{F}_i = \mathbf{F}_i^{\text{type}} + \mathbf{F}_i'. \tag{6}$$

In this article, we set the token of $\mathbf{F}_i^{\text{type}}$ to be "1."

C. Information Interacting-Enhancing Module

So far, we have obtained the textual and visual embeddings for the texts and RS images. Then, their relationships should be mined intensely for CMRSITR. Also, the current visual embeddings should be enhanced as they cannot fully represent the complex contents within RS images. To achieve these goals, we develop IIEM in this section. As shown in Fig. 1, IIEM comprises three feature interacting-enhancing (FIE) blocks, which are combined serially. Each of them consists of an intermodality relationship interacting (IMRI) subblock and a visual feature enhancing (VFE) subblock. Now, we will explain them one by one. Since the operations of the three FIE blocks are the same, we only use the first FIE block as an example to illustrate clearly.

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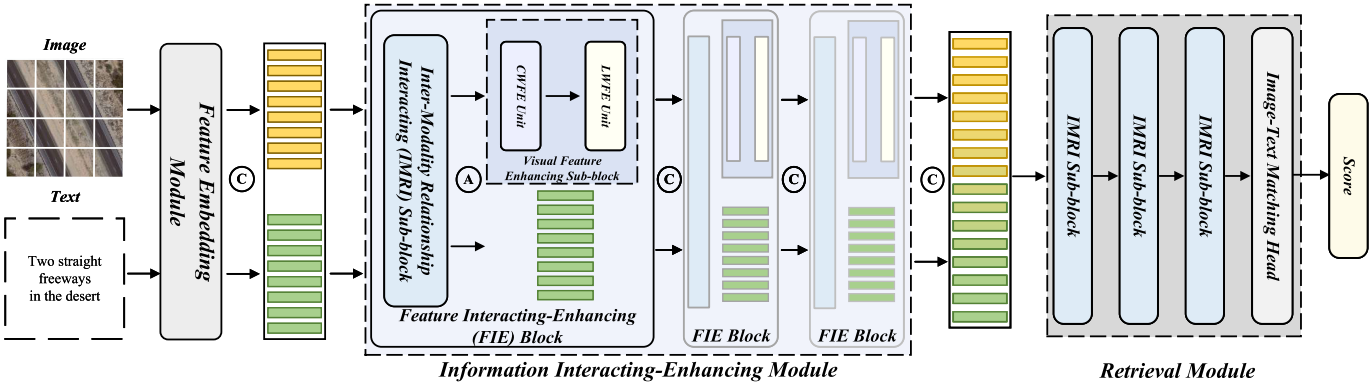


Fig. 1. Framework of the proposed IEFT. There are three main components, including an FEM, an IIEM, and an RM. FEM leverages the linear projector and Bert to transform RS images and texts into d-dimensional visual and textual feature embeddings. IIEM aims to model intermodality relationships between different features and enhance the visual feature representation simultaneously, achieved by the IMRI subblocks and VFE subblocks. RM encloses three IMRI subblocks and an ITM head, focusing on further narrowing the semantic gaps between the two modalities and outputs matching scores that reflect the similarities of RS images and texts. Here, Ⓒ means we concatenate the image and text features into a joint feature, and Ⓐ indicates that we divide the joint feature into image and text features.

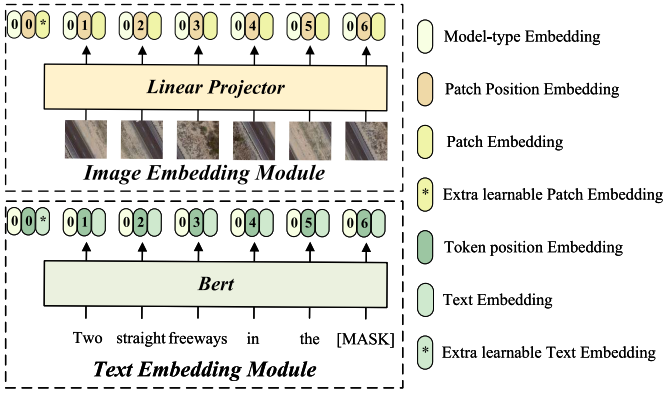


Fig. 2. Flowchart of FEM, which comprises an image embedding block and a text embedding block. The image embedding block leverages a simple linear projector to generate visual embeddings for RS images, while the text embedding block adopts Bert to map texts into textual embeddings. The different embeddings have the same dimensions, facilitating cross-modal retrieval.

1) Intermodality Relationship Interacting Subblock: To bridge the gap between the two modalities, we use a transformer-encoder as our IMRI subblock, which contains an MHA layer, two layer-norm layers, and a multilayer perception (MLP). The residual connection integrates these components. As shown in Fig. 3, the MHA aims at modeling the inherent relationships between texts and RS images (spatialwise), while the layer-norm layers and MLP focus on guaranteeing the IMRI's stability and improving its nonlinear fitness capacity.

For an input data $\mathbf{X}_i$=[$\mathbf{G}_i$, $\mathbf{F}_i$] $\in \mathbb{R}^{d\times(l+n+2)}$, it would be normalized and passed through the MHA layer that mainly has h SAM in parallel. SAM can leverage the vectors' dependence attention matrix to model the relationships between text and RS image embeddings. It is accomplished by the scale dot-production, whose formulation is

$$\mathrm{SAM}\left(\mathbf{X}_i^{\mathrm{ln}}\right)=\mathrm{softmax}\left(\frac{\mathbf{E}^{Q}\left(\mathbf{X}_i^{\mathrm{ln}}\right)\left(\mathbf{E}^{K}\left(\mathbf{X}_i^{\mathrm{ln}}\right)\right)^{T}}{\sqrt{d_k}}\right)\left(\mathbf{E}^{V}\left(\mathbf{X}_i^{\mathrm{ln}}\right)\right) \tag{7}$$

where $\mathbf{X}_i^{\mathrm{ln}}$ specifies the normalized format of the input data that is accomplished by the layer normalization operation LN($\cdot$), $\mathbf{E}^{Q}(\cdot)$, $\mathbf{E}^{K}(\cdot)$, and $\mathbf{E}^{V}(\cdot)$ denote three projection matrixes that map the normalized input data $\mathbf{X}_i^{\mathrm{ln}}$ into queries, keys, and values, d_k indicates a scaled factor, and softmax($\cdot$) is the softmax function. After h parallel SAMs, $\mathbf{X}_i$ is transformed into $\{\mathbf{S}_i^1,\ldots,\mathbf{S}_i^h\}$. Then, by contacting them together and applying a linear operation, the output of the MHA layer can be obtained. This step can be expressed as

$$\mathrm{MHA}\left(\mathbf{X}_i^{\mathrm{ln}}\right)=\mathrm{concat}\left(\mathbf{S}_i^1,\ldots,\mathbf{S}_i^h\right)\mathbf{E}^{o} \tag{8}$$

where concat($\cdot$) implies the contact operation and $\mathbf{E}^{o}$ means the linear mapping. Next, the summation of MHA($\mathbf{X}_i^{\mathrm{ln}}$) and $\mathbf{X}_i$ would be fed into the second layer-norm layer and MLP. Here, the MLP contains two fully connected layers connected by a Gaussian error linear unit (GeLU).

As a whole, for the input data $\mathbf{X}_i$, the process IMRI subblock can be formulated as

$$\begin{aligned}\mathbf{X}_i^{\mathrm{ln}}&=\mathrm{LN}(\mathbf{X}_i)\\ \mathbf{X}_i^{\mathrm{mha}}&=\mathbf{X}_i+\mathrm{MHA}\left(\mathbf{X}_i^{\mathrm{ln}}\right)\\ \mathbf{X}_i^{\mathrm{imri}}&=\mathbf{X}_i^{\mathrm{mha}}+\mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{X}_i^{\mathrm{mha}}\right)\right)\end{aligned} \tag{9}$$

where $\mathbf{X}_i^{\mathrm{mha}} \in \mathbb{R}^{d\times(l+n+2)}$ indicates the output of the MHA layer, MLP($\cdot$) means the operations of MLP, and $\mathbf{X}_i^{\mathrm{imri}} \in \mathbb{R}^{d\times(l+n+2)}$ denotes the output of the IMRI subblock.

2) Visual Feature-Enhancing Subblock: To enhance the visual embeddings, we first split the visual component $\mathbf{Z}_i^{v} \in \mathbb{R}^{d\times n}$ from $\mathbf{X}_i^{\mathrm{imri}}$ according to the image-type identification. The rest part is recorded $\mathbf{Z}_i^{o} \in \mathbb{R}^{d\times(l+2)}$. Then, it would be fed into a channelwise feature-enhancing (CWFE) unit and a localwise feature-enhancing (LWFE) unit orderly (see Fig. 1).

Like the proposed IMRI subblock, the base of the CWFE unit (see Fig. 4) is also the transformer-encoder which focuses on exploring the inner relationships hidden in the input data. Nevertheless, the CWFE unit tends to capture the helpful knowledge within RS images from the channel perspective. Particularly, $\mathbf{Z}_i^{v}$ is first transformed into the channelwise feature matrix $\mathbf{Z}_i^{vT} \in \mathbb{R}^{n\times d}$ by the transpose operation. Then,

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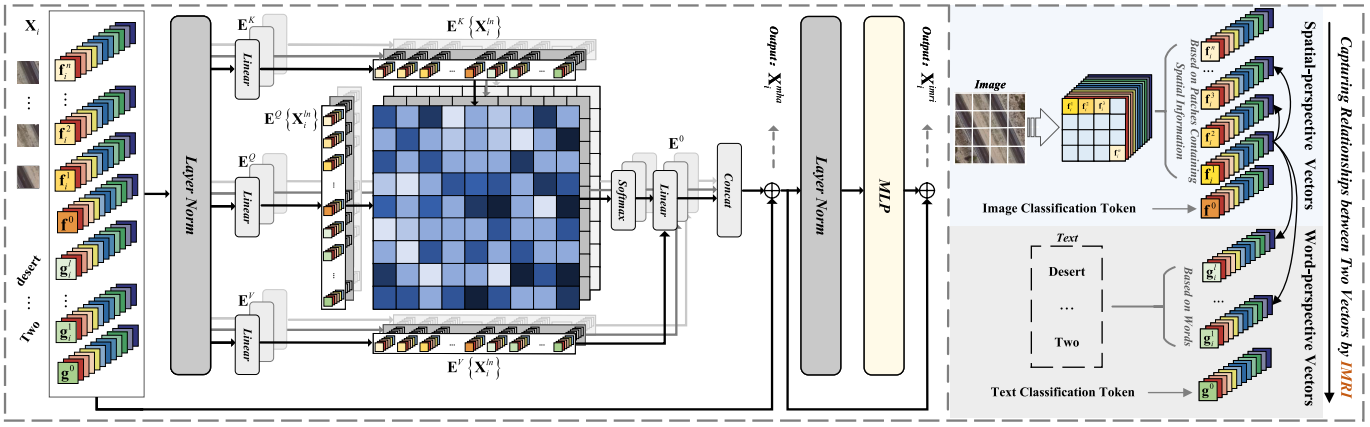


Fig. 3. (Left) Structure of the IMRI subblock, which incorporates the SAM to model the inherent relationships between the RS image and text modalities. (Right) Schematic illustration of the goals of the IMRI subblock. The input data $\mathbf{X}_i$ is concatenated by the textual embeddings $\mathbf{G}_i = [\mathbf{g}^0, \mathbf{g}_i^1, \ldots, \mathbf{g}_i^l]$ and the visual embeddings $\mathbf{F}_i = [\mathbf{f}^0, \mathbf{f}_i^1, \ldots, \mathbf{f}_i^n]$. Here, we record each item in $\mathbf{G}_i$ as a word-perspective vector, which represents a word in this text. Besides, each element within $\mathbf{F}_i$ is regarded as a spatial-perspective vector, which describes a patch obtained from the RS image. After passing $\mathbf{X}_i$ through the IMRI subblock, the relations between word- and spatial-perspective vectors can be captured by SAM.

it would be passed through a normalization layer and a h-head SAM to perform an attention mechanism for channelwise relationships. The definitions of SAM can refer to (7) and (8). Next, another normalization layer and an MLP layer are applied. Note that the above two steps are connected in a residual way. The above process can be formulated as

$$\begin{aligned}\mathbf{Z}_i^{\mathrm{ln}T} &= \mathrm{LN}\left(\mathbf{Z}_i^{vT}\right)\\ \mathbf{Z}_i^{\mathrm{mha}T} &= \mathbf{Z}_i^{vT} + \mathrm{MHA}\left(\mathbf{Z}_i^{\mathrm{ln}T}\right)\\ \mathbf{Z}_i^{\mathrm{cwfe}T} &= \mathbf{Z}_i^{\mathrm{mha}T} + \mathrm{MLP}\left(\mathrm{LN}\left(\mathbf{Z}_i^{\mathrm{mha}T}\right)\right)\end{aligned} \tag{10}$$

where $\mathbf{Z}_i^{\mathrm{ln}T} \in \mathbb{R}^{n\times d}$ denotes the normalized feature matrix, $\mathbf{Z}_i^{\mathrm{mha}T} \in \mathbb{R}^{n\times d}$ means the channelwise output of the MHA layer, and $\mathbf{Z}_i^{\mathrm{cwfe}T} \in \mathbb{R}^{n\times d}$ means the output of the CWFE unit.

The CWFE unit pays attention to exploiting the RS images' contents from the global aspect. To discover the salient local regions and enrich the visual embeddings, the LWFE unit with the depthwise convolution [59] is developed here. In detail, we first apply a transpose operation on $\mathbf{Z}_i^{\mathrm{cwfe}T}$ to obtain $\mathbf{Z}_i^{\mathrm{cwfe}} \in \mathbb{R}^{d\times n}$. After passing through a normalization layer, we reshape it into the 2-D version $\mathbf{Z}_i^{2-\mathrm{D}} \in \mathbb{R}^{d\times\sqrt{n}\times\sqrt{n}}$. Then, two 1×1 convolutions are adopted to map $\mathbf{Z}_i^{2-\mathrm{D}}$ into $\mathbf{C}_i^1 \in \mathbb{R}^{d_u\times\sqrt{n}\times\sqrt{n}}$ and $\mathbf{C}_i^2 \in \mathbb{R}^{d_u\times\sqrt{n}\times\sqrt{n}}$ in parallel. We formulate this step as

$$\begin{aligned}\mathbf{C}_i^1 &= \mathbf{E}_1^{\mathrm{up}}\left(\mathbf{Z}_i^{2-\mathrm{D}}\right)\\ \mathbf{C}_i^2 &= \mathbf{E}_2^{\mathrm{up}}\left(\mathbf{Z}_i^{2-\mathrm{D}}\right)\end{aligned} \tag{11}$$

where $\mathbf{E}_1^{\mathrm{up}}(\cdot)$ and $\mathbf{E}_2^{\mathrm{up}}(\cdot)$ denote two 1×1 convolutions. Note that "up" means $d_u > d$. Next, we want to find the salient local areas hidden in RS images. To achieve this goal, depthwise convolution is adopted, which is good at capturing the local structure of images by learning neighborhood information. We apply a 3×3 depthwise convolution and a GeLU nonlinear activation to $\mathbf{C}_i^1$ for depressing the nonsalient backgrounds. Meanwhile, another 3×3 depthwise convolution is conducted on $\mathbf{C}_i^2$ to highlight the salient regions. By producing them in a dotted manner, we can obtain the fused feature $\mathbf{C}_i \in \mathbb{R}^{d\times\sqrt{n}\times\sqrt{n}}$, containing luxuriant local clues. The above operations can be formulated as

$$\mathbf{C}_i = \phi\left(\mathbf{W}_1^d\left(\mathbf{C}_i^1\right)\right) \times \mathbf{W}_2^d\left(\mathbf{C}_i^2\right) \tag{12}$$

where $\mathbf{W}_1^d(\cdot)$ and $\mathbf{W}_2^d(\cdot)$ mean two 3×3 depthwise convolutions, and $\phi(\cdot)$ represents the GeLU nonlinearity. Finally, a 1×1 convolution is used to reduce the channel numbers of $\mathbf{C}_i$ and the output of the LWFE unit is

$$\mathbf{C}_i^{\mathrm{out}} = \mathbf{E}^{\mathrm{down}}(\mathbf{C}_i) + \mathbf{Z}_i^{2-\mathrm{D}} \tag{13}$$

where $\mathbf{C}_i^{\mathrm{out}} \in \mathbb{R}^{d\times\sqrt{n}\times\sqrt{n}}$.

After reshaping $\mathbf{C}_i^{\mathrm{out}}$ into $\mathbf{Z}_i^{\mathrm{out}} \in \mathbb{R}^{d\times n}$ and contacting $\mathbf{Z}_i^{\mathrm{out}}$ and $\mathbf{Z}_i^o$, the contextualized feature $\mathbf{X}_i^{\mathrm{out}(1)} \in \mathbf{R}^{d\times(n+l+2)}$ of the first FIE block can be generated. The other two FIE blocks operate the same. In this way, we can get the output of IIEM $\mathbf{X}_i^{\mathrm{out}}$. Overall, the LWFE leverages the depthwise convolution to capture localwise information and finally enhances our model's ability to discriminate nonsalient backgrounds and salient regions.

D. Retrieval Module

So far, we have obtained $\mathbf{X}_i^{\mathrm{out}}$, which is rich in intermodality relationships and RS image contents. Now, RM is developed to accomplish cross-modal retrieval.

As shown in Fig. 1, RM consists of three IMRI subblocks and an ITM head. Their functions are further to explore the relations between RS images and texts and to generate the matching scores $\mathbf{S}_i \in [0, 1]$ that are used to measure the similarities between RS images and texts. The larger scores, the greater similarities between the RS images and texts. The structure and process of the IMRI subblock can be found in Section III-C1. After three IMRI subblocks, $\mathbf{X}_i^{\mathrm{out}}$ can be transformed into $\mathbf{X}_i^f \in \mathbb{R}^{d\times(l+n+2)}$. The ITM head consists of a linear mapping $\mathbf{E}^{\mathrm{itm}}(\cdot)$, a nonlinear activation layer $\tanh(\cdot)$, and a matching score mapping $\mathbf{E}^{\mathrm{match}}(\cdot)$. The matching scores can be obtained by

$$\mathbf{S}_i = \mathbf{E}^{\mathrm{match}}\left(\tanh\left(\mathbf{E}^{\mathrm{itm}}\left(\mathbf{X}_i^f\right)\right)\right). \tag{14}$$

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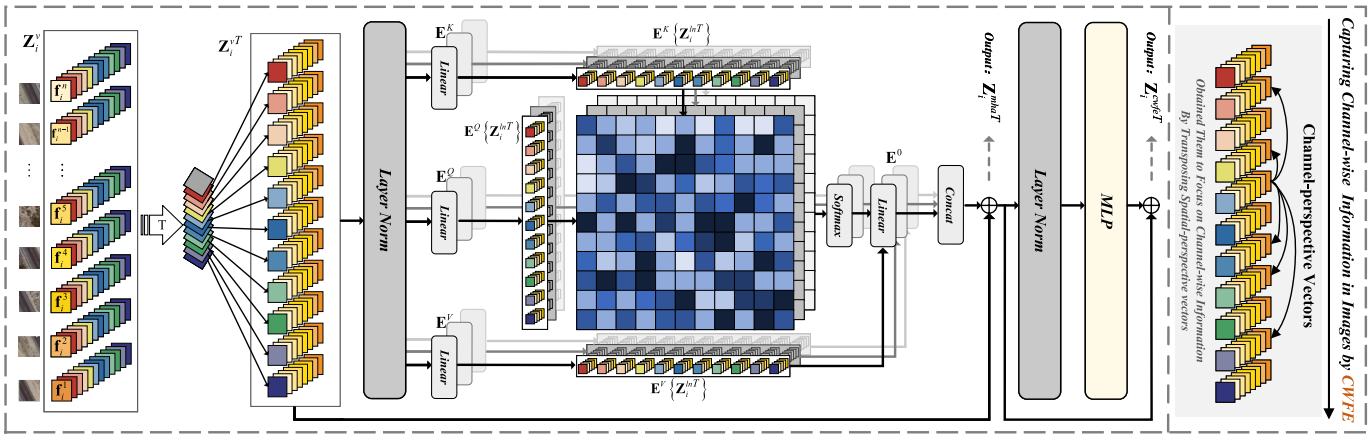


Fig. 4. (Left) Structure of the CWFE unit, which tends to capture the helpful knowledge within RS images from the channel aspect. (Right) Schematic illustration of the targets of the CWFE unit. When the visual component $\mathbf{Z}_i^v$ is input, we first transform it into $\mathbf{Z}_i^{vT}$. Here, $\mathbf{Z}_i^v$ represents the RS images from the spatial aspect as it is obtained by the RS patches. Thus, $\mathbf{Z}_i^{vT}$ reflects knowledge of the RS images from the channel dimension. Then, SAM is applied on $\mathbf{Z}_i^{vT}$ to capture channelwise relationships hidden in RS images.

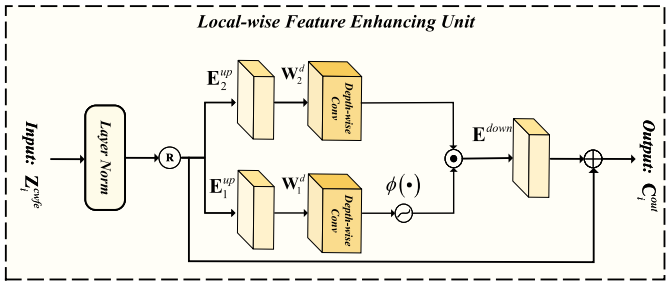


Fig. 5. Flowchart of the LWFE unit, which mainly incorporates a normalization layer, three 1×1 convolutions, two 3×3 depthwise convolutions, and a GeLU activation function. The function of this unit is to enhance the visual representation by mining salient local regions within RS images.

To train our IEFT using the obtained matching scores, we first construct positive and negative samples for each RS image. In particular, we treat an RS image and its paired texts as positive samples. At the same time, the RS image and λ random texts are regarded as negative samples. Then, the cross-entropy loss function is selected to minimize the gaps between $\mathbf{S}_i$ and the relationship label $\mathbf{m}$, that is,

$$\mathcal{L}_{\text{matching}} = \text{CrossEntropy}(\mathbf{S}_i, \mathbf{m}). \tag{15}$$

In the testing stage, we first count the matching scores between the query (an RS image or a text) and all candidates (texts or RS images in the archive). Then, the retrieval list can be determined in descending order of scores.

IV. Experiments and Discussion

A. Testing Datasets Introduction

Four popular RS image datasets are employed to evaluate our IEFT, including the RSICD [60], RSITMD [16], UCM-Captions [61], and Sydney-Captions [61] datasets.

The RSICD dataset is a large-scale RS image caption dataset, containing 10 921 RS images with varied pixel resolutions and fixed sizes (224×224). These RS images are divided into 30 categories, and the number of RS images per class varies from 240 to 1031. Also, each RS image has five description sentences in RSICD. RSITMD is a derivate of RSICD, with 4743 256×256 RS images. Parts are selected from the RSICD dataset, and Google Earth collects others. There are 32 classes in this dataset, and the numbers of RS images within each class are imbalanced. Like RSICD, each RS image in RSITMD has five textual captions. The UCM-Captions dataset was constructed based on the UC Merced Land Use dataset [62]. UCM-Captions contains 2100 RS images and are divided into 21 semantic classes equally. Their pixel resolutions and sizes are 1 ft and 256×256, respectively. Five texts are manually annotated to describe a single RS image. It is noted that the descriptions for the RS images from the same category are similar [60], which enhances the difficulties of CMRSITR. The Sydney-Captions dataset was released in [61], and the original version is the Sydney dataset [63]. This dataset contains 613 RS images with the size of 500×500, and every image has five captions. Each image's spatial resolution is approximately 0.5 m, and all RS images are divided into seven semantic categories. The detailed information related to RS images' semantic categories and their volumes of four datasets can be found in the second section of the Supplementary Material.

B. Experimental Setting

1) Implementation Details: All experiments are carried out on two NVIDIA RTXA6000 GPUs with 48 GB memory. According to the JSON documents of four datasets, we set the ratios of the training and testing sets are 90% and 10% for different archives, respectively. In the training phase, we adopt the AdamW optimizer with the initialized learning rate of 10^{-4} and weight decay of 10^{-2} to update the parameters of our IEFT. The batch size equals 8 for all datasets, and the epochs are 100 (Sydney-Captions and UCM-Captions) and 50 (RSICD and RSITMD), respectively. The learning rate is warmed up at the beginning 10% of epochs, then decays to zero linearly. The CWFE and LWFE units are initialized randomly, and the ViLT's pretrained parameters initialize the other parts of our IEFT [64], [65], [66], [67]. Note that we

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TABLE I
LENGTH OF TEXTS ACROSS FOUR DATASETS

Data Sets	Average Length	Maximum Length
RSICD [60]	10.55	34
RSITMD [16]	10.25	34
UCM-Captions [61]	10.51	22
Sydney-Captions [61]	12.21	20

repeat the experiments five times, and the exhibited results in this article are the average values.

For texts, we first splice the [cls] and [sep] at the beginning and end of each text, respectively. Then, the lengths of texts are normalized to 40, that is, $l + 1 = 40$. The reasons for fixing the text length of 40 are twofold. The first one is the specification of the pretrained parameters of ViLT [23]. The ViLT is pretrained on millions of online image–text pairs data, providing rich cross-modal information that can be leveraged to enhance our model's results. To keep consistency with ViLT, we set the text length to 40. The second one is the consideration of textual information integrity. After analyzing the texts in four testing datasets (see Table I), we can find that the length of texts varies from 20 to 34, and the average length equals approximately 10. Based on these observations, fixing the text length of 40 can guarantee the integrity of the textual information. To normalize all texts' length as 40 for our model, we pad "0" at the end of the textual sequences. Here, the "0" token has no semantic meaning. Besides, since the predefined vocabulary (in Section III-B1) has 30 522 frequent words [68], the dimension of texts' one-hot codes are 30 522, that is, $d_o = 30\,522$.

For RS images, we first resize each RS image into 384×384 and slice it into $n = 144$ patches with the size of 32×32. Then, the RandAugment [69] is applied to augmented RS images without the color inversion and cutout. In our IEFT, the dimensions of RS images and texts are varied at different stages, and we set them $d = 768$ and $d_u = 2042$, respectively. The numbers of SAMs in the IMRI subblock and the CWFE unit are 12, that is, $h = 12$. Besides, the kernel size of depthwise convolutions in the LWFE unit is set as 3.

2) Evaluation Metrics: To confirm the effectiveness of our IEFT, we choose two widely used evaluation metrics as $R@k$ (where $k = 1, 5, 10$) and mR [10]. $R@k$ is the ratio of ground truth appearing in the retrieval results of top k. Note that the ground truth shows the relationship label between the image or text with the query. The relationship labels equal to 1 means that the RS images and texts are positive sample pairs. Otherwise, they are regarded as negative sample pairs. To obtain the $R@k$, we should first obtain k image–text relationship labels corresponding to the top k matching scores between a query with all retrieval entities. Then we need to identify how many positive samples appear in the top k retrieval samples. Thus, the $R@k$ can be defined as

$$R@k = \frac{1}{B}\sum_{i=1}^{B} \text{Precision}(i, k) \tag{16}$$

where B means the volume of the query set, i means the ith query set, and $\text{Precision}(i, k)$ denotes whether the positive items corresponding to ith query appear in the top k retrieval samples or not. If the positive sample exists in the top k retrieval results, $\text{Precision}(i, k) = 1$. Otherwise, $\text{Precision}(i, k) = 0$. Furthermore, mR is the average of all $R@k$s.

When we leverage the above assessment criteria to judge the retrieval models, we found that the issue of duplicated texts (a textual caption may appear in different RS images) would impact them negatively. As shown in Fig. 6(a), there are two RS images and their corresponding texts. It is easy to find that the caption texts are duplicated. Logically speaking, when we use "This is a dense forest" as the query, these two RS images are both the correct "Image retrieval results"; or when we use one of the two RS images as the query, the "This is a dense forest" should be the correct "Text retrieval results." However, each text links an "Image Label" in the dataset so that the sentence "This is a dense forest" will be changed into two sentences "This is a dense forest. (Image label1)" and "This is a dense forest. (Image label2)" When we use "This is a dense forest. (Image label1)" as a query, the second RS is regarded as an incorrect result. Similarly, when we use the first RS image as the query, the " This is a dense forest. (Image label2)" would be deemed as an improper retrieval result. The schematic illustration of this phenomenon can be found in Fig. 6(b), which is somewhat unreasonable and disturbs the credibility of $R@k$s and mR.

To overcome this dilemma, we develop a simple strategy to improve the robustness of assessment criteria. In detail, we first unify the texts in the dataset without duplication. Then, the "Image Labels" corresponding to each text are counted. Thus, one text may have several "Image Labels," such as "This is a dense forest: Related Image labels:1, 2." Consequently, when we use the above sentence as the query, the first and second RS images are both the correct "Image retrieval results." Also, this sentence can be the right "Text retrieval result" for two RS images. This process can be found in Fig. 6(c). In this way, the obtained $R@k$s and mR values are more objective. For clarity, we record two indicators $IMR@k$ and $IMmR$ in this article. In the following experiments, we use $IMR@k$ and $IMmR$ to count the numerical results. The Supplementary Material will also summarize the original $R@k$ and mR values for reference.

C. Sensitivity Analysis

As mentioned in Section III-D, we should set the number of negative texts λ before training the proposed IEFT. Thus, in this section, we study its influence on IEFT and provide the optimal values for different datasets. To this end, we vary the values of λ from 11 to 19 with an interval of 2. Based on different λ, we can train diverse IEFTs, and their performance for four archives is displayed in Fig. 7. By observing them, we can find that: 1) all IEFTs' behavior is close to each other; 2) the values of the assessment criteria are enhanced gradually at first and then descends to a degree; and 3) the peak values for different datasets appear at $\lambda = 15$. The above investigation confirms that introducing the negative texts into training is necessary, which can make IEFT keep semantic consistency reasonably. According to the contents discussed above, we set $\lambda = 15$ for the following experiments.

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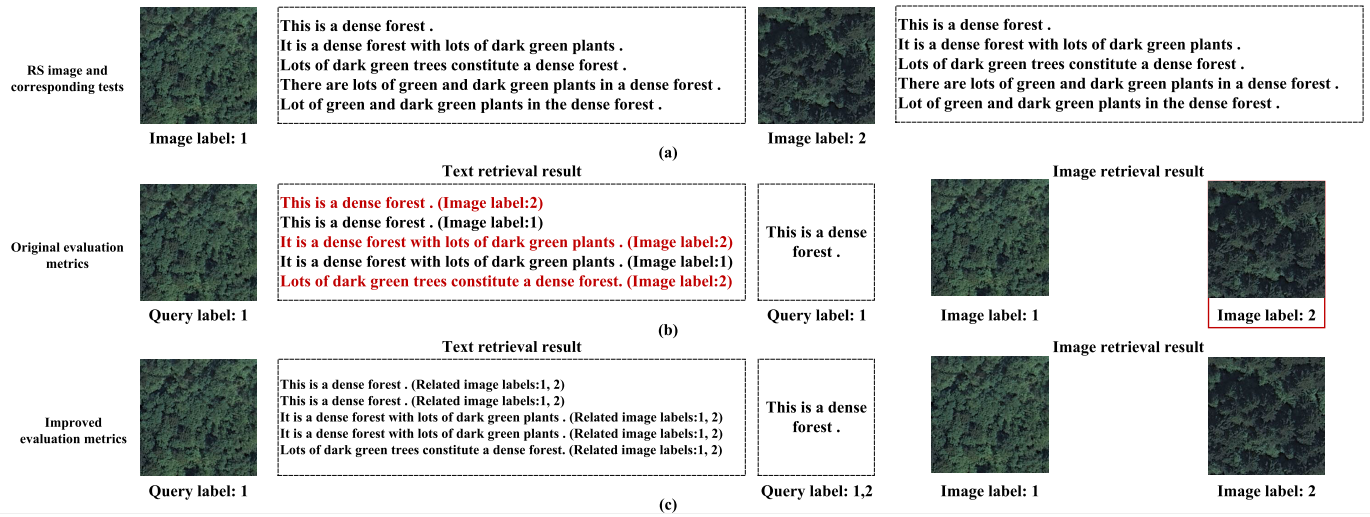


Fig. 6. Examples illustrate the negative influence of duplicated texts and the operation of $IMR@k$ and $IMmR$. Here, (a) denotes two RS images with their corresponding texts, (b) shows how duplicated texts negatively affect the results, and (c) illustrates the function of improved $R@k$ and mR. Note that the uncorrected results are marked in red.

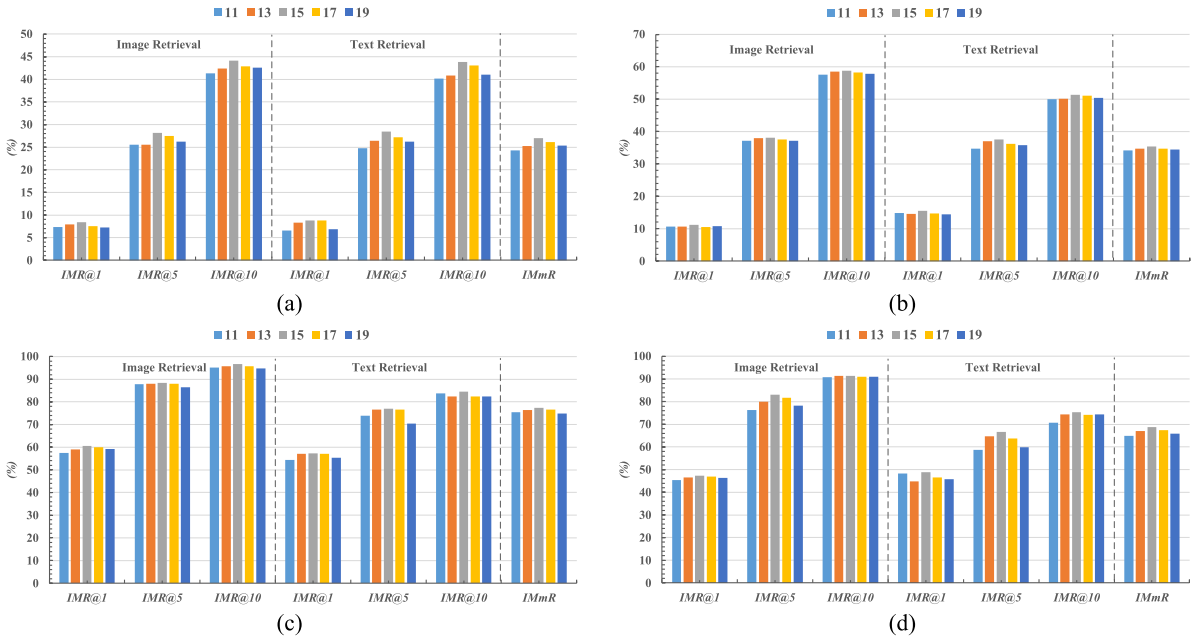


Fig. 7. Illustration of influence of parameter λ on $IMR@k$s and $IMmR$s of four datasets. Here, (a) is the results of the RSICD dataset, (b) is the RSITMD dataset, (c) is the UCM-Caption dataset, and (d) is the Sydney-Captions dataset.

D. Ablation Study

As exhibited in Fig. 1, our IEFT involves an FEM, an IIEM, and an RM. Among the three modules, IIEM is the most complex one. It consists of three FIE blocks, each of which mainly contains an IMRI subblock and a VFE subblock. The IMRI subblock aims at modeling the relationships between RS images and texts by SAM. The VFE subblock focuses on enhancing the RS images' representations by exploring the channelwise and localwise knowledge hidden in images, which is achieved by the CWFE and LWFE units. We construct the following four networks to study their contributions to the proposed IEFT further.

1) *Net-1:* FEM + RM.
2) *Net-2:* FEM + three IMRIs + RM.
3) *Net-3:* FEM + (three IMRIs + CWFEs) + RM.
4) *Net-4:* FEM + (three IMRIs + CWFEs + LWFEs) + RM.

Here, Net-1 only comprises an FEM and an RM. Based on Net-1, Net-2 adds three IMRI subblocks between FEM and RM. Then, Net-3 embeds the CWFE units into three IMRI subblocks. Finally, Net-4 introduces the LWFE units on the basis of Net-3. Actually, Net-4 is our IEFT. The performance of the four networks is displayed in Table II. Here, "Text Retrieval" denotes that we use RS images as the queries to search the matched textual captions. While "Image Retrieval" means the opposite scenario. Note that the experimental settings are the same as the contents discussed in Section IV-B1.

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TABLE II
$IMR@k$ AND $IMmR$ VALUES (%) OF FOUR ABLATION NETWORKS COUNTED ON DIFFERENT DATASETS

	RSICD Date Set							RSITMD Data Set						
Approach	Image Retrieval			Text Retrieval			$IMmR$	Image Retrieval			Text Retrieval			$IMmR$
	$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$		$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$	
Net-1	5.00	18.34	30.69	3.75	16.92	26.71	16.91	6.64	23.75	39.44	6.85	22.78	34.29	22.29
Net-2	7.02	26.72	42.24	7.95	27.38	42.84	25.69	10.75	36.94	57.52	13.64	34.50	48.60	33.65
Net-3	7.59	27.63	43.53	8.13	27.90	43.03	26.30	11.10	37.88	58.41	15.33	34.80	50.73	34.70
Net-4	**8.38**	**28.17**	**44.16**	**8.78**	**28.47**	**43.88**	**26.97**	**11.19**	**38.09**	**58.84**	**15.49**	**37.61**	**51.40**	**35.43**
	UCM-Captions Data Set							Sydney-Captions Data Set						
Approach	Image Retrieval			Text Retrieval			$IMmR$	Image Retrieval			Text Retrieval			$IMmR$
	$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$		$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$	
Net-1	49.95	77.51	86.86	31.90	50.47	61.42	59.68	38.66	73.97	87.73	31.03	50.50	62.06	57.33
Net-2	56.44	87.68	95.15	51.94	71.44	82.66	74.22	43.36	80.54	90.94	43.09	59.19	66.66	63.96
Net-3	60.43	88.08	96.48	53.36	71.58	82.69	75.43	45.82	81.28	91.40	46.20	62.81	72.50	66.67
Net-4	**60.59**	**88.29**	**96.61**	**57.29**	**76.92**	**84.60**	**77.38**	**47.33**	**83.14**	**91.45**	**48.85**	**66.66**	**75.28**	**68.78**

We can easily find that the values of all assessment criteria are increased from Net-1 to Net-4. For example, compared with Net-1, the significant enhancements in $IMmR$ achieved by Net-4 are 10.06% (RSICD), 13.14% (RSITMD), 17.70% (UCM-Captions), and 11.45% (Sydney-Captions). Taking the "Image Retrieval" results of RSITMD as another example, the improvements in $IMR@10$ obtained by Net-4 are 19.4% (Net-1), 1.32% (Net-2), and 0.43% (Net-3), respectively. The above discussion confirms the contributions of the IMRI subblocks, CWFE units, and LWFE units to our IEFT. Furthermore, we can summarize three points from the results.

1) The distinct performance gaps between Net-1 and Net-2 indicate that the IMRI subblocks can effectively model the relationships between RS images and texts. The acceptance results of different CMRSITR tasks can be generated based on those useful relationships. This also implies the importance of relationship creation in the cross-modal scenario.
2) After introducing the CWFE unit into Net-2, the behavior of Net-3 becomes stronger than that of Net-2. This illustrates that the explored channelwise information is helpful in RS image understanding. By supplementing the channelwise clues into the global features, the obtained visual features are beneficial to describe the complex contents within RS images and to establish visual–textual relations.
3) When we add LWFE into Net-3, Net-4 produces the best results, which demonstrates the usefulness of exploiting the local information from RS images. Not only can the discrimination of visual features be enhanced in this way, but also the rich knowledge hidden in RS images can be discovered. Therefore, we can get more credible visual and textual relations for CMRSITR tasks.

To further study the effectiveness of the IMRI subblock, we conduct the following experiments. In particular, we vary the number of IMRI subblocks from 1 to 5 based on Net-2, and the results counted on the four datasets are shown in Fig. 8. By observing them, we can find that: 1) the $IMmR$ values consistently rise as the number of IMRI subblocks increases and 2) the most substantial improvements appear when the number of the IMRI subblock changes from 2 to 3. These imply the positive contributions of the IMRI subblock. Also, the more IMRI subblocks, the stronger behavior of the final network. Nevertheless, the cost of good performance is a significant increase in time complexity. Considering the tradeoff between the performance and time costs, we suggest the number of the IMRI subblocks equals three.

E. Comparisons With Other Approaches

To comprehensively evaluate the retrieval performance of the proposed IEFT, we compare it with the following models.

1) *CLIP [70]:* This model adopts the specific encoders to map the different input data into a multimodal embedding space for image–text search tasks. The adopted image encoders are ResNet50 [18] and ViT-B/32 [26], and the selected text encoder is a 12-layer transformer [24]. We use ViT-B/32 and the 12-layer transformer in the following experiments to complete the data embedding and cross-modal retrieval.
2) *MSIT [11]:* After obtaining the embeddings of RS images and the corresponding textual captions, multiscale interactive transformer (MSIT) directly constructs a single-branch transformer-based network to learn the semantic relationships between them for CMRSITR. Also, a lightweight multiscale feature exploration submodule is developed and embedded in MSIT to mine the important local information hidden in RS images.
3) *AMFMN [16]:* AMFMN is a dual-branch network that uses a DCNN to learn RS image features and a gate recurrent unit (GRU) model to obtain text features. Then, the visual-guided attention (VGA) module is developed to narrow the semantic gap between visual and textual features so that credible CMRSITR results can be generated. AMFMN provides three different attention operations, that is, soft attention, fusion attention, and similarity attention. In the following experiments, we record AMFMN with different attention ways AMFMN-soft, AMFMN-fusion, and AMFMN-sim, respectively.
4) *LW-MCR [8]:* Like AMFMN, a lightweight multiscale crossmodal text-image retrieval method (LW-MCR) also leverages a DCNN to learn the visual features from RS images. Unlike AMFMN, LW-MCR adopts the group convolution scheme to obtain the textual features. Then, LW-MCR introduces knowledge distillation to achieve satisfactory and efficient CMRSITR results. According to the knowledge distillation manner, two models (LW-MCR-b and LW-MCR-d) are used in the following.

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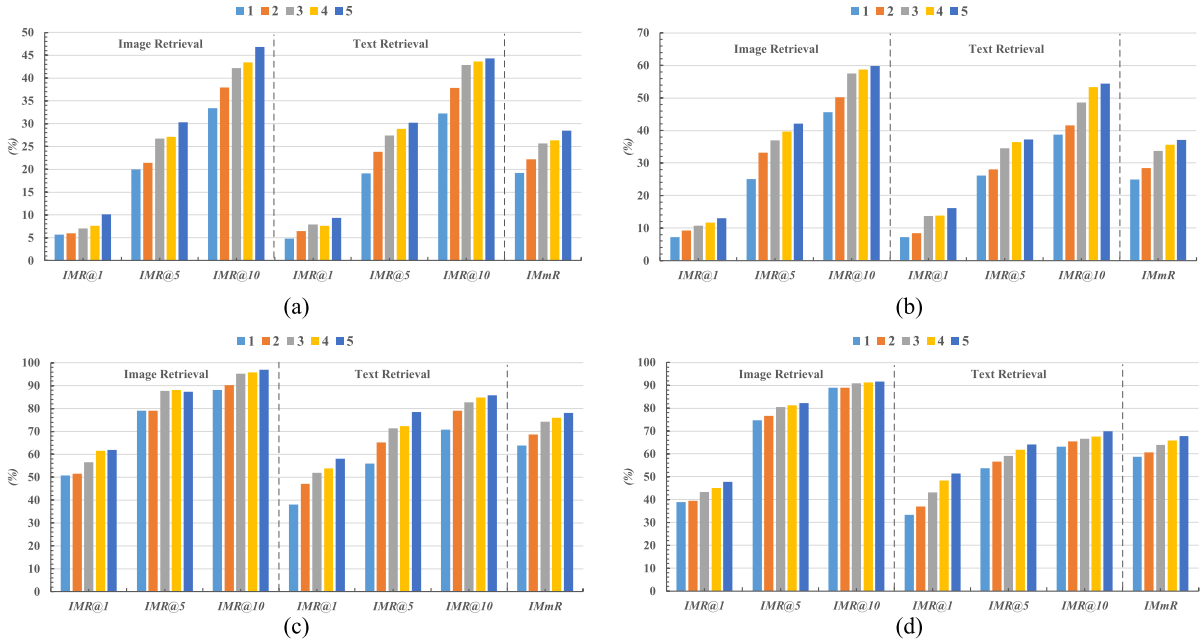


Fig. 8. $IMR@k$s and $IMmR$s values of Net-2 with different numbers of the IMRI subblocks. (a) RSICD. (b) RSITMD. (c) UCM-Captions. (d) Sydney-Captions.

LW-MCR-b means that there is no other network during the knowledge distillation, and LW-MCR-d indicates a teacher network exists in the knowledge distillation.

5) *GaLR [10]:* A novel retrieval framework based on global and local information (GaLR) first proposes a multilevel information dynamic fusion (MIDF) scheme to explore both the global and local information within RS images and texts. Then, a multivariate rerank algorithm is developed to refine the cross-modal retrieval results.

Here, contrastive language-image pre-training (CLIP) and MSIT are transformer-based methods, and others are DCNN-based methods. We simulate CLIP and MSIT by ourselves, and the DCNN-based methods are accomplished by the released codes. All free parameters are set according to the original literature, and the experimental settings are consistent with ours. Among all compared methods, MSIT, LW-MCR, and GaLR have some assistive designs or technologies. Hence, we make some changes when we complete them in the following experiments for fairness, which are given in detail.

1) MSIT has six transformer encoders in its main module to learn the relations between RS images and texts. However, IEFT only has three transformer encoders (located in IIEM). Accordingly, we reduce the number of transformer encoders in MSIT to three.
2) In LW-MCR, apart from the cross-modal learning network with a knowledge distillation strategy, the authors introduce a large number of unlabeled RS images collected from Google Earth to enhance the retrieval performance. Nevertheless, the proposed IEFT does not have any auxiliary information. Thus, in the following experiments, we remove the unlabeled data learning part from the main body of LW-MCR.
3) To explore the helpful information from RS images, GaLR adopts object detection techniques. In addition, a specific reranking scheme is added after the initial retrieval to improve the final accuracy. Nonetheless, the above techniques do not exist in the proposed IEFT. Therefore, we discard object detection and image reranking when we complete GaLR.

1) Results of the RSICD Dataset: The RSICD dataset is the largest among the four datasets, and the results of different methods are shown in Table III. From the observation of the results, we can find that all methods' performance is acceptable, and the behavior of the transformer-based methods (CLIP, MSIT, and our IEFT) is stronger than that of the DCNN-based models. This implies that the mechanism of the transformer is beneficial to exploit the intermodality relationships between RS images and texts in RSICD, leading to the small semantic gaps between the visual and textual features and better CMRSITR results. Although the compared methods achieve good results, the proposed IEFT outperforms them in all cases. Taking $IMmR$ as examples, the enhancements achieved by IEFT are 1.99% (over CLIP), 3.42% (over MSIT), 6.01% (over AMFMN-soft), 5.98% (over AMFMN-fusion), 5.67% (over AMFMN-sim), 8.75% (over LW-WCR-b), 7.70% (over LW-MCR-d), and 5.87% (over GaLR). The reasons behind this encouraging investigation are twofold. First, IEFT processes RS image and text embeddings simultaneously, which can mitigate the misaligned problem in multimodal feature distributions. Second, owing to the specific structure of the developed IIEM, not only the relations between RS images and texts, but also the rich information hidden in RS images can be explored. The positive results discussed above confirm the usefulness of IEFT for the RSICD dataset.

2) Results of the RSITMD Dataset: The CMRSITR results of different methods are summarized in Table III. Like RSICD, our method achieves the best performance on the RSITMD dataset. For example, in the "Image Retrieval" scenario, the $IMR@10$ values of IEFT is 58.54%, whose

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TABLE III
$IMR@k$ AND $IMmR$ VALUES (%) OF DIFFERENT COMPARISON METHODS COUNTED ON DIFFERENT DATASETS

Approach	RSICD Date Set							RSITMD Data Set						
	Image Retrieval			Text Retrieval			$IMmR$	Image Retrieval			Text Retrieval			$IMmR$
	$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$		$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$	
CLIP [70]	8.30	26.77	41.97	7.08	25.88	39.93	24.98	8.23	29.24	44.48	7.51	24.11	36.50	25.01
MSIT [11]	6.53	24.78	39.92	6.70	24.27	39.09	23.55	9.34	37.71	58.83	14.37	33.18	48.74	33.70
AMFMN-soft [16]	6.82	23.30	35.72	8.13	20.94	30.83	20.96	10.70	36.56	55.45	12.53	31.11	44.09	31.74
AMFMN-fusion [16]	6.61	22.77	35.22	9.39	20.61	31.34	20.99	10.82	36.17	53.24	13.12	30.82	42.91	31.18
AMFMN-sim [16]	7.51	23.49	36.72	8.39	21.22	30.49	21.30	10.05	36.57	56.16	12.68	30.52	43.03	31.50
LW-WCR-b [8]	4.70	17.77	31.25	4.75	19.76	31.10	18.22	9.69	34.73	54.33	12.83	27.21	37.38	29.36
LW-WCR-d [8]	6.47	20.91	34.98	6.31	18.38	28.54	19.27	10.70	34.69	50.39	13.49	28.98	39.60	29.64
GaLR [10]	6.43	23.64	37.18	8.35	20.49	30.49	21.10	10.95	35.19	53.21	13.12	30.45	42.92	30.97
IEFT	**8.38**	**28.17**	**44.16**	**8.78**	**28.47**	**43.88**	**26.97**	**11.19**	**38.09**	**58.84**	**15.49**	**37.61**	**51.40**	**35.43**
Approach	UCM-Captions Data Set							Sydney-Captions Data Set						
	Image Retrieval			Text Retrieval			$IMmR$	Image Retrieval			Text Retrieval			$IMmR$
	$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$		$IMR@1$	$IMR@5$	$IMR@10$	$IMR@1$	$IMR@5$	$IMR@10$	
CLIP [70]	43.77	72.00	82.91	32.85	45.55	54.59	55.28	44.72	77.31	84.63	40.79	55.74	66.08	61.55
MSIT [11]	51.64	84.54	92.60	49.20	68.09	78.56	70.77	44.03	74.56	86.20	48.09	64.10	74.32	65.23
AMFMN-soft [16]	49.45	82.74	94.62	45.13	64.67	77.47	69.01	38.19	82.11	91.39	43.30	61.15	70.79	64.49
AMFMN-fusion [16]	49.98	83.29	94.68	44.94	64.48	78.27	69.27	43.22	81.60	90.45	42.15	61.04	68.21	64.45
AMFMN-sim [16]	51.34	83.92	95.23	49.60	67.06	78.56	70.95	42.75	82.06	89.31	39.65	60.34	74.13	64.71
LW-WCR-b [8]	46.48	79.20	92.14	38.98	59.22	72.61	64.77	43.44	80.68	90.68	37.93	51.72	70.68	62.52
LW-WCR-d [8]	52.19	79.52	88.09	45.71	57.61	67.61	65.12	37.91	82.89	91.18	44.82	62.06	72.41	65.22
GaLR [10]	47.45	75.96	85.64	39.36	51.42	60.31	60.02	41.03	80.11	88.38	31.03	54.01	72.41	61.16
IEFT	**60.59**	**88.29**	**96.61**	**57.29**	**76.92**	**84.60**	**77.38**	**47.33**	**83.14**	**91.45**	**48.85**	**66.66**	**75.28**	**68.78**

improvements compared with other methods are 14.36% (over CLIP), 0.01% (over MSIT), 3.39% (over AMFMN-soft), 5.60% (over AMFMN-fusion), 2.68% (over AMFMN-sim), 4.51% (over LW-WCR-b), 8.45% (over LW-MCR-d), and 5.63% (over GaLR). For another example, in the "Text Retrieval" scenario, the $IMR@10$ value of IEFT is 51.40%, whose enhancements in comparison to other methods are 14.90% (over CLIP), 2.66% (over MSIT), 7.31% (over AMFMN-soft), 8.49% (over AMFMN-fusion), 8.37% (over AMFMN-sim), 14.02% (over LW-WCR-b), 11.80% (over LW-MCR-d), and 8.48% (over GaLR). These promising results illustrate that our IEFT is helpful to the RSITMD dataset. A notable observation is that the behavior of CLIP is the weakest. The main reasons behind this phenomenon are as follows. First, CLIP is developed for natural images and ignores the characteristics of RS images (such as multiscale). Second, since the volume of samples in RSITMD is not big enough, the pretrained parameters of CLIP cannot be optimized to the ideal state. This significant change implies the importance of considering RS scenes' specific characteristics.

3) Results of the UCM-Captions Dataset: The diverse methods' CMRSITR results are exhibited in Table III. Similar to the results of RSICD and RSITMD, IEFT is the best among all methods. For $IMmR$, our IEFT achieves 77.38%, which is 22.10%, 6.61%, 8.37%, 8.11%, 6.43%, 12.61%, 12.26%, and 17.36% higher than CLIP, MSIT, AMFMN-soft, AMFMN-fusion, AMFMN-sim, LW-WCR-b, LW-WCR-d, and GaLR, respectively. Also, the biggest improvements of $IMR@10$ are as high as 14.36% and 14.90% in the "Image Retrieval" and "Text Retrieval" cases, respectively. Those observations confirm the effectiveness of the proposed IEFT again. We are pleasantly surprised to note that the enhancements achieved by our method on the UCM dataset are more distinct than RSICD and RSITMD. This is largely due to the following reasons. On the one hand, compared with the other two datasets, the number of RS images per category in the UCM dataset is balanced, which is helpful to our IEFT sufficiently modeling relationships between RS images and their corresponding texts. On the other hand, besides exploring the relations between the different data, IEFT focuses on capturing the complex contents within RS from different aspects, which is conducive to CMRSITR tasks directly.

4) Results of the Sydney-Captions Dataset: The results of various methods are exhibited in Table III. Not surprisingly, the performance of IEFT ranks first among all methods. For "Image Retrieval," the $IMR@k$ values of IEFT are approximately 1.05 ($IMR@1$), 1.01 ($IMR@5$), and 1.00 ($IMR@10$) times as many as that of the second places (CLIP, LW-WCR-d, and LW-WCR-b). Similarly, for "Text Retrieval," the $IMR@k$ values of IEFT are approximately 1.01 ($IMR@1$), 1.04 ($IMR@5$), and 1.01 ($IMR@10$) times as many as that of the second places (MIST). As a whole, compared to the second place (MSIT), the $IMmR$ value IEFT is 1.05 times as many as that of MSIT. These positive results confirm the superiorities of our IEFT again.

Apart from the above numerical results, we also: 1) study the IEFT's performance across the different categories in different datasets; 2) analyze the IEFT's behavior under the noisy scenario (i.e., there are some negative words in the caption sentences); and 3) provide visual retrieval examples. From observing those results, we can find that our IEFT can still perform well. Detailed information can be found in the Supplementary Material.

F. Time Cost Discussion

In this section, we study the efficiency of our method by counting its inference times and parameter size.

For inference times, we randomly select an RS image (or a text) to be the query and retrieve the corresponding texts (or RS images) from the testing set using the trained IEFT. The results (averaged values of ten random experiments) are exhibited in Table IV. The inference times of the other seven compared methods are also reported for reference. Note that the experimental settings of different methods are the same

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TABLE IV
RETRIEVAL EFFICIENCY OF DIFFERENT METHODS (UNIT: SECOND)

Approach	RSICD	RSITMD	UCM-Captions	Sydney-Captions
CLIP [70]	2.1072	1.1758	0.5809	0.1551
MSIT [11]	2.2359	1.2573	0.6238	0.1896
AMFMN-soft [16]	0.0704	0.0274	0.0165	0.0054
AMFMN-fusion [16]	0.0977	0.0284	0.0199	0.0056
AMFMN-sim [16]	0.0679	0.0235	0.0158	0.0051
LW-MCR [8]	**0.0438**	**0.0143**	**0.0876**	**0.0023**
GaLR [10]	0.0632	0.0225	0.0161	0.0042
IEFT	5.2599	2.9591	1.4660	0.4310

TABLE V
PARAMETERS VOLUME OF DIFFERENT METHODS (UNIT: MILLIONS)

Methods	CLIP [70]	MSIT [11]	AMFMN [16]	LW-WCR [8]	GaLR [10]	IEFT (Ours)
Params	151.28	90.32	35.94	1.65	73.31	100.12

as Table III. By observing those results, we can find that the inference times of all methods stay within a reasonable range. Nevertheless, due to the MHA mechanism, the time costs of the transform-based methods (CLIP, MSIT, and IEFT) are longer than that of DCNN-based models (AMFMN-soft, AMFMN-fusion, AMFMN-sim, LW-MCR, and GaLR). The most time-saving model is LW-MCR, while our IEFT is the most time-consuming model. The reason behind this investigation can be summarized as follows. The essential part of our IEFT is the transformer encoder, which contains multiple SAMs to explore the relations between RS images and texts. The time complexity of these SAMs is high, so the time costs of the proposed network are relatively high. Fortunately, the distinct performance gaps between IEFT and other methods (especially the DCNN-based models) illustrate that the extra time is well spent.

The parameter volume of different models is displayed in Table V. By observing the results, we find that the transformer-based models (CLIP, MSIT, and IEFT) are heavier than the DCNN-based methods (AMFMN, LW-WCR, and GaLR). This is because the transformer encoder has more parameters compared with components within DCNN. Besides, IEFT is lighter than CLIP but heavier than MSIT. The reasons are that CLIP has more transformer encoders than IEFT, and IEFT's feature learning module (i.e., IIEM) is larger than MSIT's feature extract part. This investigation means the proposed IEFT is not the largest one, at least. In addition, the parameter sizes of different parts within our IEFT are 25.83 million (FEM), 52.32 million (IIEM), and 21.97 million (RM).

A notable observation is that although IEFT is not the heaviest model, its inference times are the longest. This is due to the CWFE unit embedded in IIEM. As mentioned in Section III-C2, CWFE tends to capture the helpful knowledge within RS images from the channel aspect. To this end, it would transpose the input feature map and then apply SAM. Since the number of channels is always large, the computational complexity of CWFE would be expensive. The possible solutions for this limitation are: 1) using sparse learning [71], [72] to optimize the attention mechanism; 2) decreasing the length of input tokens gradually [73]; and 3) altering the attention mechanism by the spatial reduction attention (such as PVTv2 [74]). Solving this shortcoming and balancing our model's inference times and performance are our future works.

V. CONCLUSION

This article proposes an IEFT model to accomplish the CMRSITR task. It regards the RS images and texts as a whole and processes them simultaneously, which can effectively learn multimodel features and directly align their cross-modal semantic relationships. After mapping RS images and texts into the feature embeddings, IEFT first uses IIEM to deeply study the cross-modal data's similarity relations (IMRI). Also, the visual features are enhanced in IIEM through a channelwise process (CWFE) and a localwise operation (LWFE). Then, the RM is developed and embedded at the top of our model to model the resemblance relationships further and produce the matching scores. Finally, the cross-modal retrieval results can be decided by those scores. The positive experimental results counted on four public datasets demonstrate the effectiveness of IEFT.

Although the IEFT's performance is well, there is still room for improvement. First, the IEFT only emphasizes the intermodal similarities or differences when it studies the cross-modal relationships. However, the intermodal affinity is ignored. Another deficiency of IEFT is the relatively high time complexity. Our future work will focus on these two points to optimize.

REFERENCES

[1] Y. Gu, Y. Wang, and Y. Li, "A survey on deep learning-driven remote sensing image scene understanding: Scene classification, scene retrieval and scene-guided object detection," *Appl. Sci.*, vol. 9, no. 10, p. 2110, May 2019.

[2] G. Cheng, X. Xie, J. Han, L. Guo, and G. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

[3] I. Kotaridis and M. Lazaridou, "Remote sensing image segmentation advances: A meta-analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 309–322, Mar. 2021.

[4] Y. Li, J. Ma, and Y. Zhang, "Image retrieval from remote sensing big data: A survey," *Inf. Fusion*, vol. 67, pp. 94–115, Mar. 2021.

[5] X. Tang, X. Zhang, F. Liu, and L. Jiao, "Unsupervised deep feature learning for remote sensing image retrieval," *Remote Sens.*, vol. 10, no. 8, p. 1243, Aug. 2018.

[6] C. Liu, J. Ma, X. Tang, F. Liu, X. Zhang, and L. Jiao, "Deep hash learning for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3420–3443, Apr. 2021.

[7] X. Tang et al., "Meta-hashing for remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2021.

[8] Z. Yuan et al., "A lightweight multi-scale crossmodal text-image retrieval method in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–19, 2021.

[9] Q. Cheng, Y. Zhou, P. Fu, Y. Xu, and L. Zhang, "A deep semantic alignment network for the cross-modal image-text retrieval in remote sensing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4284–4297, 2021.

[10] Z. Yuan et al., "Remote sensing cross-modal text-image retrieval based on global and local information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620616.

[11] Y. Wang, J. Ma, M. Li, X. Tang, X. Han, and L. Jiao, "Multi-scale interactive transformer for remote sensing cross-modal image-text retrieval," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 839–842.

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[12] X. Tang et al., "An unsupervised remote sensing change detection method based on multiscale graph convolutional network and metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[13] Y. Yang et al., "AR^{2}Det: An accurate and real-time rotational one-stage ship detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[14] X. Tang et al., "Hyperspectral image classification based on 3-D octave convolution with spatial–spectral attention network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2430–2447, Mar. 2021.

[15] Y. Liu, Q. Li, Y. Yuan, Q. Du, and Q. Wang, "ABNet: Adaptive balanced network for multiscale object detection in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[16] Z. Yuan et al., "Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4404119.

[17] Y. Lv, W. Xiong, X. Zhang, and Y. Cui, "Fusion-based correlation learning model for cross-modal remote sensing image retrieval," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[19] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.

[20] F. Faghri, D. J. Fleet, J. R. Kiros, and S. Fidler, "VSE++: Improving visual-semantic embeddings with hard negatives," 2017, *arXiv:1707.05612*.

[21] C. Feichtenhofer et al., "Masked autoencoders as spatiotemporal learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 35, 2022, pp. 35946–35958.

[22] X. Li et al., "OSCAR: Object-semantics aligned pre-training for vision-language tasks," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2020, pp. 121–137.

[23] W. Kim, B. Son, and I. Kim, "ViLT: Vision-and-language transformer without convolution or region supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 5583–5594.

[24] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[25] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[26] A. Dosovitskiy et al., "An image is worth 16 × 16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[27] B. Kim, J. Lee, J. Kang, E. Kim, and H. J. Kim, "HOTR: End-to-end human-object interaction detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 74–83.

[28] N. Wang, W. Zhou, J. Wang, and H. Li, "Transformer meets tracker: Exploiting temporal context for robust visual tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1571–1580.

[29] Z. Jiang, Z. Xiang, J. Xu, and M. Zhao, "LGT-Net: Indoor panoramic room layout estimation with geometry-aware transformer network," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1644–1653.

[30] Y.-C. Chen et al., "UNITER: Universal image-text image representation learning," in *Proc. 16th Eur. Conf. Comput. Vis. (ECCV)*. Glasgow, U.K.: Springer, Aug. 2020, pp. 104–120.

[31] S. Li, T. Xiao, H. Li, W. Yang, and X. Wang, "Identity-aware textual-visual matching with latent co-attention," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1908–1917.

[32] J. Gu, J. Cai, S. Joty, L. Niu, and G. Wang, "Look, imagine and match: Improving textual-visual cross-modal retrieval with generative models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7181–7189.

[33] M. Żelaszczyk and J. Mańdziuk, "Cross-modal text and visual generation: A systematic review. Part 1: Image to text," *Inf. Fusion*, vol. 93, pp. 302–329, May 2023.

[34] Q. You, J. Luo, and Z. Zhang, "End-to-end convolutional semantic embeddings," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 5735–5744.

[35] L. Zhen, P. Hu, X. Wang, and D. Peng, "Deep supervised cross-modal retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 10386–10395.

[36] Z. Zheng, L. Zheng, M. Garrett, Y. Yang, M. Xu, and Y.-D. Shen, "Dual-path convolutional image-text embeddings with instance loss," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 16, no. 2, pp. 1–23, May 2020.

[37] C. Wang, L. Li, C. Yan, Z. Wang, Y. Sun, and J. Zhang, "Cross-modal semantic correlation learning by Bi-CNN network," *IET Image Process.*, vol. 15, no. 14, pp. 3674–3684, Dec. 2021.

[38] Z. Ji, H. Wang, J. Han, and Y. Pang, "SMAN: Stacked multimodal attention network for cross-modal image–text retrieval," *IEEE Trans. Cybern.*, vol. 52, no. 2, pp. 1086–1097, Feb. 2022.

[39] K.-H. Lee, X. Chen, G. Hua, H. Hu, and X. He, "Stacked cross attention for image-text matching," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 201–216.

[40] Y. Huang, Q. Wu, C. Song, and L. Wang, "Learning semantic concepts and order for image and sentence matching," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 6163–6171.

[41] K. Li, Y. Zhang, K. Li, Y. Li, and Y. Fu, "Visual semantic reasoning for image-text matching," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4653–4661.

[42] X. Xu, T. Wang, Y. Yang, L. Zuo, F. Shen, and H. T. Shen, "Cross-modal attention with semantic consistence for image–text matching," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 12, pp. 5412–5425, Dec. 2020.

[43] K. Li, Y. Zhang, K. Li, Y. Li, and Y. Fu, "Image-text embedding learning via visual and textual semantic reasoning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 641–656, Jan. 2023.

[44] J. Lu, D. Batra, D. Parikh, and S. Lee, "ViLBERT: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–11.

[45] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 34, 2021, pp. 9694–9705.

[46] G. Mikriukov, M. Ravanbakhsh, and B. Demir, "Unsupervised contrastive hashing for cross-modal retrieval in remote sensing," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 4463–4467.

[47] A. Radford et al., "Improving language understanding by generative pre-training," Tech. Rep., 2018.

[48] A. Radford et al., "Language models are unsupervised multitask learners," *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.

[49] T. B. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neur. Inf. Process. Sys.*, vol. 33, 2020, pp. 1877–1901.

[50] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surveys*, vol. 54, no. 10s, pp. 1–41, Jan. 2022.

[51] Y. Bazi, L. Bashmal, M. M. A. Rahhal, R. A. Dayil, and N. A. Ajlan, "Vision transformers for remote sensing image classification," *Remote Sens.*, vol. 13, no. 3, p. 516, Feb. 2021.

[52] P. Lv, W. Wu, Y. Zhong, F. Du, and L. Zhang, "SCViT: A spatial-channel feature preserving vision transformer for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409512.

[53] J. Ma, M. Li, X. Tang, X. Zhang, F. Liu, and L. Jiao, "Homo–heterogenous transformer learning framework for RS scene classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2223–2239, 2022.

[54] X. Tang, M. Li, J. Ma, X. Zhang, F. Liu, and L. Jiao, "EMTCAL: Efficient multiscale transformer and cross-level attention learning for remote sensing scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5626915.

[55] Q. Wang, Y. Liu, Z. Xiong, and Y. Yuan, "Hybrid feature aligned network for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624915.

[56] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[57] Q. Li, R. Zhong, X. Du, and Y. Du, "TransUNetCD: A hybrid transformer network for change detection in optical remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622519.

[58] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313.

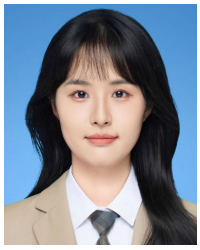
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[59] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M. Yang, "Restormer: Efficient transformer for high-resolution image restoration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 5718–5729.
[60] X. Lu, B. Wang, X. Zheng, and X. Li, "Exploring models and data for remote sensing image caption generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 2183–2195, Apr. 2018.
[61] B. Qu, X. Li, D. Tao, and X. Lu, "Deep semantic understanding of high resolution remote sensing image," in *Proc. Int. Conf. Comput., Inf. Telecommun. Syst. (CITS)*, Jul. 2016, pp. 1–5.
[62] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geographic Inf. Syst.*, Nov. 2010, pp. 270–279.
[63] F. Zhang, B. Du, and L. Zhang, "Saliency-guided unsupervised feature learning for scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2175–2184, Apr. 2015.
[64] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2014, pp. 740–755.
[65] R. Krishna et al., "Visual genome: Connecting language and vision using crowdsourced dense image annotations," *Int. J. Comput. Vis.*, vol. 123, no. 1, pp. 32–73, May 2017.
[66] V. Ordonez, G. Kulkarni, and T. Berg, "Im2Text: Describing images using 1 million captioned photographs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 24, 2011, pp. 1–9.
[67] P. Sharma, N. Ding, S. Goodman, and R. Soricut, "Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning," in *Proc. 56th Annu. Meeting Assoc. Comput. Linguistics*, 2018, pp. 2556–2565.
[68] Y. Wu et al., "Google's neural machine translation system: Bridging the gap between human and machine translation," 2016, *arXiv:1609.08144*.
[69] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "Randaugment: Practical automated data augmentation with a reduced search space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 3008–3017.
[70] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
[71] R. Child, S. Gray, A. Radford, and I. Sutskever, "Generating long sequences with sparse transformers," 2019, *arXiv:1904.10509*.
[72] L. Fan et al., "Embracing single stride 3D object detector with sparse transformer," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 8448–8458.
[73] Z. Dai, G. Lai, Y. Yang, and Q. Le, "Funnel-Transformer: Filtering out sequential redundancy for efficient language processing," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, pp. 4271–4282, 2020.
[74] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, no. 3, pp. 415–424, Sep. 2022.



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